

RESEARCH INTEREST

Privacy Compliance, Natural Language Processing, Formal Methods, Machine Learning

- *Neural-symbolic reasoning, trustworthy AI, and translation between natural language and formal logic*

EDUCATION

Stony Brook University

Stony Brook, NY, USA

- *Ph.D. in Computer Science, Department of Computer Science*

Aug. 2023 – Present

- **Advisor:** Dr. Omar Chowdhury

- **Research Area:** Formal verification, Privacy compliance, and LLM-based specification synthesis

University of New Brunswick

Fredericton, Canada

- *M.Sc. in Computer Science, Faculty of Computer Science*

May 2021 – May. 2023

- **Advisor:** Dr. Ali Ghorbani

- **Thesis:** Transferability of Machine Learning Algorithms for IoT Device Profiling and Identification

Kwame Nkrumah University of Science and Technology

Kumasi, Ghana

- *B.Sc. in Computer Engineering, College of Engineering*

Sep. 2012 – Jun. 2016

- **Project:** An Integrated Messaging Platform for an Enterprise Environment

RESEARCH AND INDUSTRY EXPERIENCE

Formal Verification and Security Group, Stony Brook University

New York, USA

- *Graduate Research Assistant*

Aug. 2023 – Present

- **Advisor:** Dr. Omar Chowdhury

- Lead researcher on evaluating large language models for natural language-to-Linear Temporal Logic (LTL) translation.

- Designed multi-dimensional benchmarks to assess semantic equivalence, satisfiability, and trace-based verification.

- Designing a typed intermediate representation for translating natural language into FOL for theorem proving

- Currently formalizing privacy compliance (HIPAA, GDPR, SOX, GLBA) using logic-based frameworks and LLM agentic reasoning.

Canadian Institute for Cybersecurity, University of New Brunswick

Fredericton, Canada

- *Graduate Research Assistant*

May. 2021 – Dec. 2022

- **Advisor:** Dr. Ali Ghorbani

- Conducted applied machine learning research on IoT device profiling and intrusion detection.

- Built and released realistic IoT profiling datasets and ensemble-based detection pipelines.

PEER-REVIEWED CONFERENCE PUBLICATIONS

(* denotes equal contribution)

- [C1] **Danso, P.K.**, et al., “ATHENA: Answering Regulatory Permissibility Questions through Iterative Fact Finding”. In *Proceedings on Privacy Enhancing Technologies (PoPETs)* (2027) Under review, Issue 2
- [C2] **Danso, P.K.**, et al., “Syntax Is Easy, Semantics Is Hard: Evaluating LLMs for LTL Translation”. In *Proceedings of the 2026 ACM Secure Development Conference (SecDev '26)* Montreal, QC, Canada 
- [C3] Dadkhah, S., Mahdikhani, H., **Danso, P.K.***, et al., “Towards the Development of a Realistic Multidimensional IoT Profiling Dataset”. In *IEEE Privacy, Security and Trust (PST 2022)* (With over 220+ citations) 
- [C4] **Danso, P.K.**, Dadkhah, S., Neto, E.C.P., et al., “Transferability of Machine Learning Algorithms for IoT Device Profiling and Identification”. In *IEEE Internet of Things Journal (IoT-J 2023)* 
- [C5] **Danso, P.K.**, Neto, E.C.P., Dadkhah, S., et al., “Ensemble-based Intrusion Detection for Internet of Things Devices”. In *IEEE Smart Communities (HONET 2022)* 
- [C6] **Danso, P.K.**, et al., “LLM-based Anomaly Detection for Digital Substation Cybersecurity”. In *IEEE Smart Communities (HONET 2025)* (To appear December 2025) 
- [C7] **Danso, P.K.**, et al., “Human-centric Machine Learning for IoT Device Identification and Vulnerability Assessment”. *HCI International 2023* 
- [C8] Zohourian, A., Dadkhah, S., Neto, E.C.P., Mahdikhani, H., **Danso, P.K.**, et al., “IoT Zigbee Device Security: A Comprehensive Review”. *Elsevier Internet of Things Journal 2023* 

TEACHING EXPERIENCE

CSE/ISE312 Legal, Social, and Ethical Issues in Computing, <i>Stony Brook University</i>	New York
▪ <i>Attend class and monitor discussions, grading, and office hours</i>	<i>Aug. 2026 – Dec. 2026</i>
ISE331 Fundamentals of Computer Security, <i>Stony Brook University</i>	New York
▪ <i>Assisted in lab instruction, grading, and office hours</i>	<i>Jan. 2024 – Apr. 2024</i>
CSE331 Computer Security Fundamentals, <i>Stony Brook University</i>	New York
▪ <i>Collaborated on assignments, held office hours, proctored and graded</i>	<i>Aug. 2023 – Dec. 2023</i>

HONORS AND AWARDS

SREB Institute on Teaching and Mentoring	
▪ <i>Selected Participant</i>	<i>2024, 2025</i>
Thirteenth Summer School on Formal Techniques + FMiTF Bootcamp	Sponsored by NSF
▪ <i>Student travel / participation award</i>	<i>May 2024</i>
CPS-IoT Week 2024 Student Travel Award	Hong Kong (NSF-sponsored)
▪ <i>Student travel award to attend CPS-IoT Week</i>	<i>April 2024</i>
iMentor Scholarship for ACM CCS	Copenhagen, Denmark (NSF-sponsored)
▪ <i>Student travel scholarship to attend ACM CCS</i>	<i>November 2023</i>
Academic Scholarship, University of New Brunswick	Fredericton, Canada
▪ <i>Faculty of Computer Science Funding</i>	<i>May 2021</i>
Institute for Analytics and Data Science Summer School Scholarship	University of Essex
▪ <i>Student scholarship to attend summer school</i>	<i>July 2020</i>
Academic Scholarship, Newmont Ahafo Development Foundation (NADeF)	Ghana
▪ <i>Merit-based undergraduate scholarship</i>	<i>September 2016</i>

PROFESSIONAL SERVICE

Paper Reviewer	
▪ <i>IEEE Internet of Things Journal</i>	<i>2023–Present</i>
Artifact Evaluation Committee	
▪ <i>Committee Member</i>	
◦ ACM CCS	<i>2024</i>
◦ USENIX Security	<i>2025</i>
Positions Held	
▪ <i>Stony Brook University</i>	
◦ Secretary: Women in Ph.D. in Computer Science (WPhD)	<i>2024–Present</i>
◦ Mentor: Center for Inclusive Education	<i>2025–Present</i>
Mentor	
▪ <i>Stony Brook University</i>	
◦ Women in Computer Science	<i>2024–Present</i>
◦ AI3 Summer Mentorship Program	
◦ AI Innovation & Diffusion REU 2026, Trustworthy AI & NLP Research Lab	<i>Summer 2026</i>